Author Response: TriPoseFusion (Submission #802)

Method	MPJPE	PA-MPJPE	PCK@.05	PCK@.10
A. Ours, calibrated ref., 52 joints, fold-0 val, paper protocol (m)				
best single view (right)	0.191	0.066	0.31	0.43
mean fusion	0.189	0.063	0.32	0.45
submitted full model	0.189	0.063	0.31	0.45
B. Ours, calibrated ref., 5-fold cross-subject, learned baselines of A3 (m)				
mean fusion	0.149	0.044	0.34	0.50
residual head, 8K	0.128	0.040	0.37	0.55
submitted arch. retrained, 870K	0.142	0.044	0.35	0.52
C. Drive&Act, official 3D ref., test subjects, zero-shot, 10 joints (mm): rigid† / PA				
mean fusion (best single view: 29.0 / 24.0)	31.6	23.1	–	–
submitted full model	32.5	23.7	–	–

Table 1. Corrected results (†rigid = rotation+translation). Outside block C's protocol: DLT, official calibration, 14 joints, absolute: 32.5 / 29.0 mm.

To all reviewers. Auditing the full evaluation stack after R-LAYQ's Tab. 3 anomaly, we found four defects: (i) transposed rotation and norm-ratio scale in the Procrustes alignment; (ii) a zero-gradient gate-entropy term; (iii) a degenerate shoulder fallback collapsing whole frames; (iv) uncalibrated, constructed extrinsics in the triangulation reference (scale error, median 2.4×). All are fixed. The reference is rebuilt by per-sequence bundle adjustment (reprojection 9.60 → 3.31 px); each sequence's free scale is anchored to the SAM3D-predicted shoulder width (cross-sequence std 0.369 → 0.015 m); PA-MPJPE and reprojection are scale-independent. Tab. 1 gives all corrected numbers (one protocol and valid-joint mask per block).

R-LAYQ W1 — metric anomaly. Cause: (i)+(iii); a synthetic regression test recovers the 0.04 m truth (old code: 0.7–2.4 m). Corrected, same masks: canonicalized mean fusion 0.971/0.564 m on the original reference (reversal gone); calibrated: Tab. 1A.

W2 — fallback rule. Missing rule: shoulder axis of the nearest earlier valid frame (a leading invalid prefix uses the first valid frame — the only look-ahead); none valid → canonical $+x$ axis; axis-parallel guard; pseudocode in §3.3.

W3 — independent reference. No independent MoCap subset was available; we drop the absolute-accuracy claim and bound the reference by validity/reprojection statistics (19jf W1) and sensitivity (conclusions unchanged across original/calibrated). Drive&Act (A2) is cross-rig transfer evidence only — its 3D reference is itself OpenPose-triangulated.

W4 — narrow the learned-component claims. We do: submitted model ≡ mean fusion (Tab. 1A, A3); claims narrow to a calibration-free canonicalization + uniform-fusion protocol with a small clean-input gain (19jf W2), the learned parts' value being corruption robustness (A1). Efficiency (0.87M params, 6.2 ms/frame CPU) is fusion-stage only; end-to-end latency incl. SAM3D is unmeasured. Minor items (mapping, frame-rate, splits, imbalance) added; joint-limit evaluation: future work.

R-19jf W1 — reference reliability under occlusion. Joints not triangulable (≥2 views within the 40 px filter) are invalid and masked out for *every* method. Validity: head/shoulder–neck 1.00, hands 0.34; hands dominate the error (0.316 m vs. torso 0.067) and are reported separately (plus a torso-only protocol). Pipeline: manual checks on 2D inputs → automatic DLT → 40 px filter → per-sequence calibration (+ uncertainty statistics).

W2 — canonicalization mechanism. Canonicalization defines the common frame enabling calibration-free fusion of per-view outputs that live in different camera frames; a raw-vs-canonicalized MPJPE is thus not meaningful (the paper's 2.078 m raw figure measured frame mismatch, not pose error). Its measurable effect is the fusion gain over the best view: small on clean input (Tab. 1A: MPJPE −1%, PA −5%, PCK@.10 +2 pts; Drive&Act PA 24.0→23.1), redundancy under view loss (dropped view: MPJPE unchanged), and calibration-free cross-rig transfer — a modest contribution, stated so in the revision.

R-xS2o A1 — gating collapse (W1–W2). We agree the stated entropy term, if active, favors uniform weights. Its implemented gradient was exactly zero (so it did not cause the collapse); retraining without it, with the fix, and with a leave-one-view-out teacher still gave 0.3333 gates — the median pseudo-teacher's trivial optimum. We remove the term, narrow the clean-input claim (W2 confirmed) and test gating only under explicit view corruptions: with corruption-augmented training a 1.9K gated head weights a zeroed view 0.037 and degrades +1–5% (mean fusion +18%; 0 for missing views).

A2 — Drive&Act (request 2). Protocol: official split-0 test subjects, no Drive&Act training; the 10 joints shared with BODY-25; PA-MPJPE main (no hips for canonical alignment) and rigid-MPJPE (Tab. 1C). Fusion beats every single view in PA; rigid-MPJPE also carries SAM3D's per-view scale bias (default shape ≈18% too large) that averaging does not remove. Published Drive&Act MPJPE is not directly comparable (joints/metrics), though our zero-shot errors are of the same order as the supervised 22.6–30.4 mm cited. 0.412 m was the defect-(iv) artifact (W3); PCK@0.05/0.02 in Tab. 1 (M3); SAM3D details (M1).

A3 — deep-learning baselines (request 3). UPose3D (calibrated multi-view 2D keypoints + uncertainty), Learnable Triangulation (2D heatmaps + cameras) and Faster VoxelPose (feature volumes) are not input-matched to our uncalibrated 3D-keypoint interface. Input-matched references (Tab. 1B, note): direct DLT (on our data the reference itself) and learned heads on our input — LT-style confidence weighting (1.9K), UPose3D-style uncertainty-aware set-attention (28K), residual (8K), and the submitted architecture retrained under the same protocol (870K), not beating simpler heads.